Reinforcement Learning for Autonomous Robot Navigation using Q-Learning

1. Overview

This project demonstrates how an autonomous robot can learn navigation using Reinforcement Learning. A grid■world environment is used where the robot learns to move from a start position to a goal while avoiding obstacles. The robot improves its behaviour through rewards and penalties using the Q■Learning algorithm.

2. Technical Highlights

- Algorithm Used: Tabular Q■Learning
- State Space: Grid positions (row, column)
- Action Space: Up, Down, Left, Right
- Reward System: +100 for reaching goal, –1 step penalty
- Exploration Strategy: ε■greedy policy
- Value Representation: Q■table

3. Q■Learning Update Rule

```
Q(s,a) = Q(s,a) + α [ R + γ max Q(s',a') – Q(s,a) ]

Where:
Q(s,a) – Current Q value
α – Learning rate
γ – Discount factor
R – Reward received
s' – Next state
```

4. Implementation Code (Python)

```
import numpy as np
import random

grid_size = 5
actions = ['up','down','left','right']

Q = np.zeros((grid_size, grid_size, len(actions)))

alpha = 0.1
gamma = 0.9
epsilon = 0.1

goal = (4,4)

def get_reward(state):
    if state == goal:
        return 100
    return -1

def choose_action(state):
    if random.uniform(0,1) < epsilon:
        return random.choice(range(len(actions)))
    return np.argmax(Q[state[0], state[1]])
```

```
def update_q(state, action, reward, next_state):
    best_next = np.max(Q[next_state[0], next_state[1]])
    Q[state[0], state[1], action] = Q[state[0], state[1], action] +         alpha * (reward + gamma
```

5. Program Output

```
Episode 50 completed
Episode 100 completed
Episode 150 completed
Episode 200 completed
Episode 250 completed
Episode 300 completed
Episode 350 completed
Episode 400 completed

Agent successfully learned optimal path to reach the goal.
```

6. Output Screenshot

Output:

```
Grid World:
[2, 0, 0, 0, 0]
[1, 1, 1, 0, 0]
[0, 0, 0, 1, 0]
[0, 1, 0, 1, 0]
[0, 0, 0, 3, 3]

Start: (0, 0)
Goal: (4, 4)
Episode 50 copleted
Episode 100 completa
Episode 150 completed
Episode 200 completed
Episode 250 completed
Episode 300 completed
Episode 350 completed
Episode 400 completed

Learned path (no exploration):
Visualized path:
S * * * *
. X X · *
. . . X *
. * . . X
. . . X G
PS C:\Users\Pooja\Stark Protocol\Projects\P>
```

7. Results

After multiple training episodes, the agent consistently reaches the goal state while avoiding obstacles. The learned policy shows efficient navigation and demonstrates the effectiveness of reinforcement learning for autonomous robot navigation tasks.

8. Tech Stack

- Programming Language: Python 3.x
- Libraries: NumPy, Random
- Development Environment: Visual Studio Code
- AI Technique: Reinforcement Learning (Q■Learning)